Introduction to simulations: hands-on session

Exploring TNG-Cluster

by Garreth Martin and Ana Contreras-Santos

Why simulations?

- Simulations bridge theory and observations
- They provide 3D physical quantities we can’t directly observe (e.g. dark matter)
- We can study the whole evolutionary history of galaxies or galaxy clusters, rather than a fixed moment in time

Goal of this session: learn to explore simulations, access the data, select objects of interest and get their properties

Types of simulations

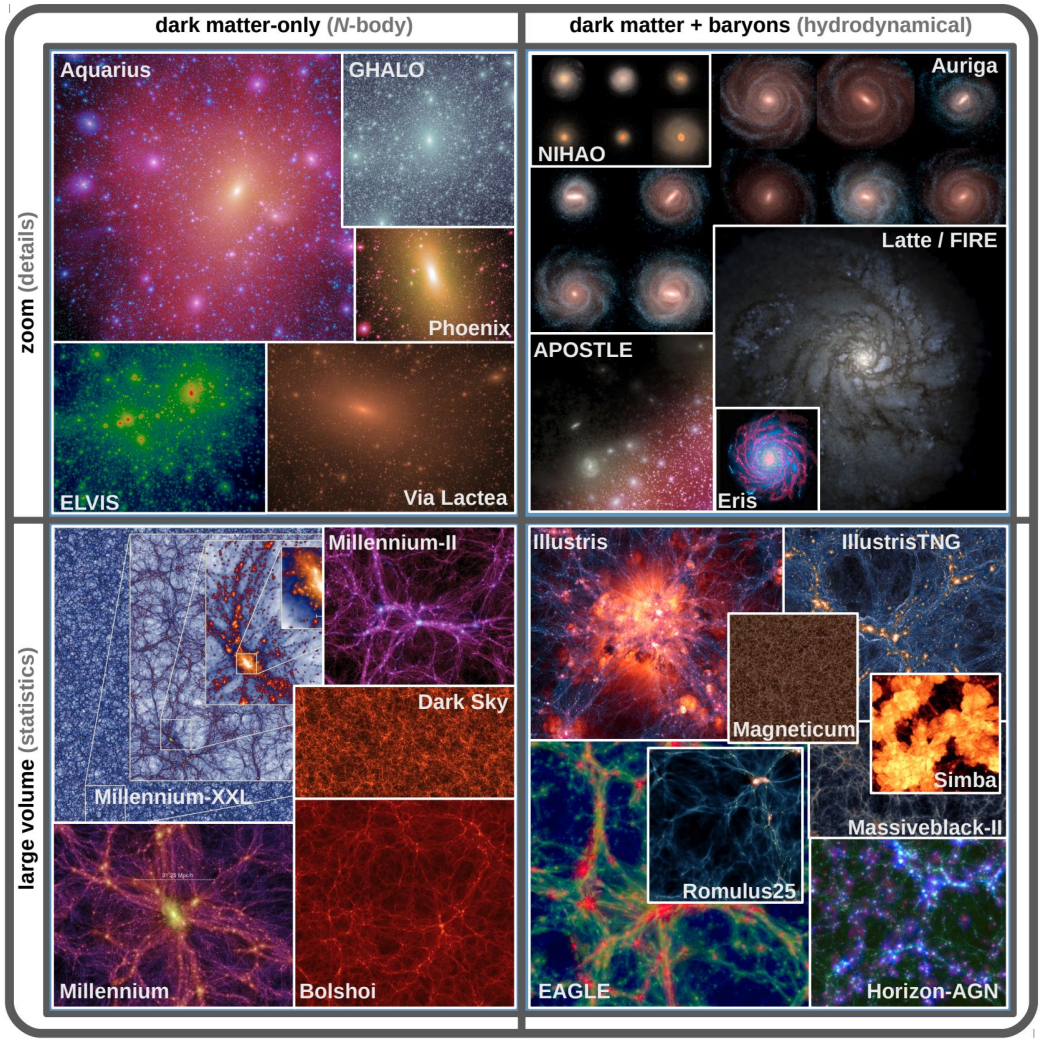
By physics included:

- Dark matter only (N-body)
- Semi-analytical models (SAMs)
- Hydrodynamical

  Numerical method:
  - SPH
  - AMR
  - Moving mesh

By simulation scale:

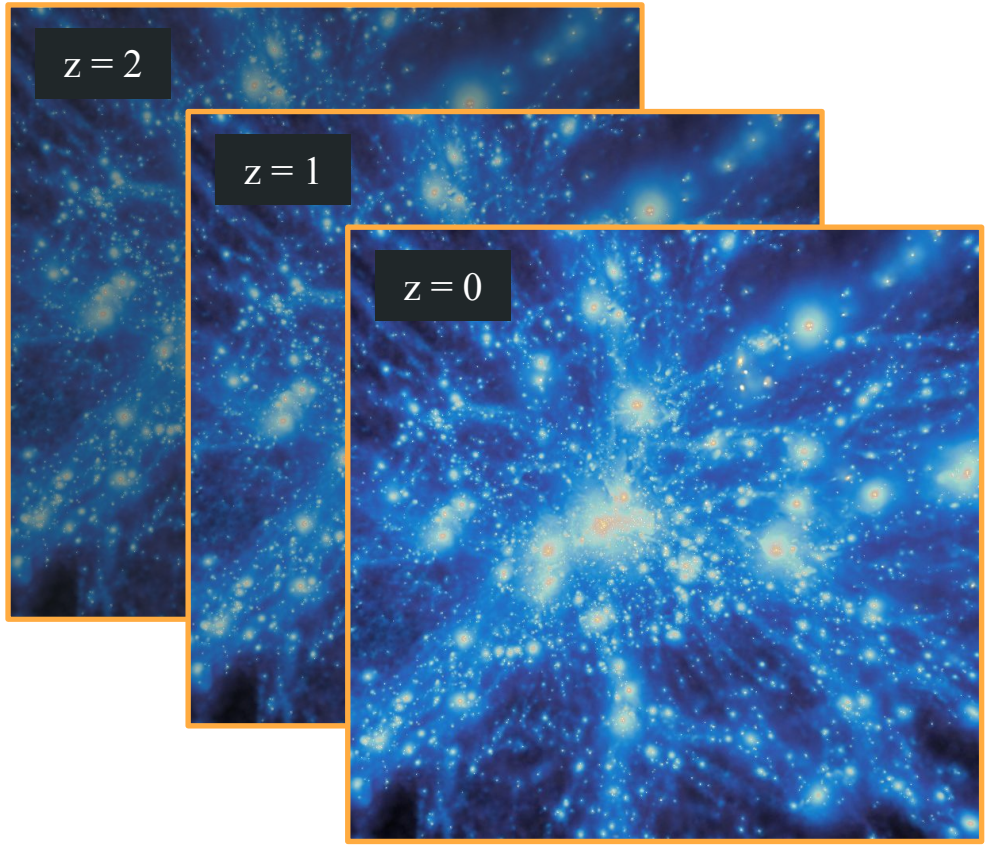
- Large boxes
- Zoom-ins



Credits: Vogelsberger et al. (2019)

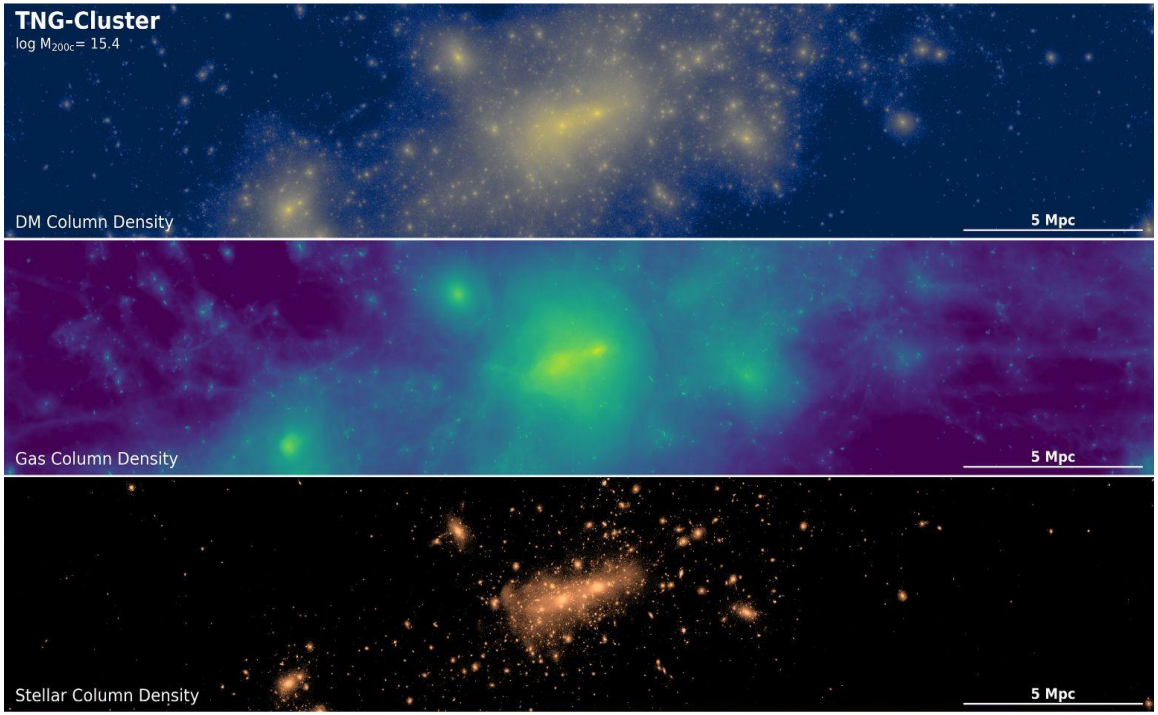
Raw output

- **Full snapshots:** 3D "photographs" of the universe at different times
- Particle data



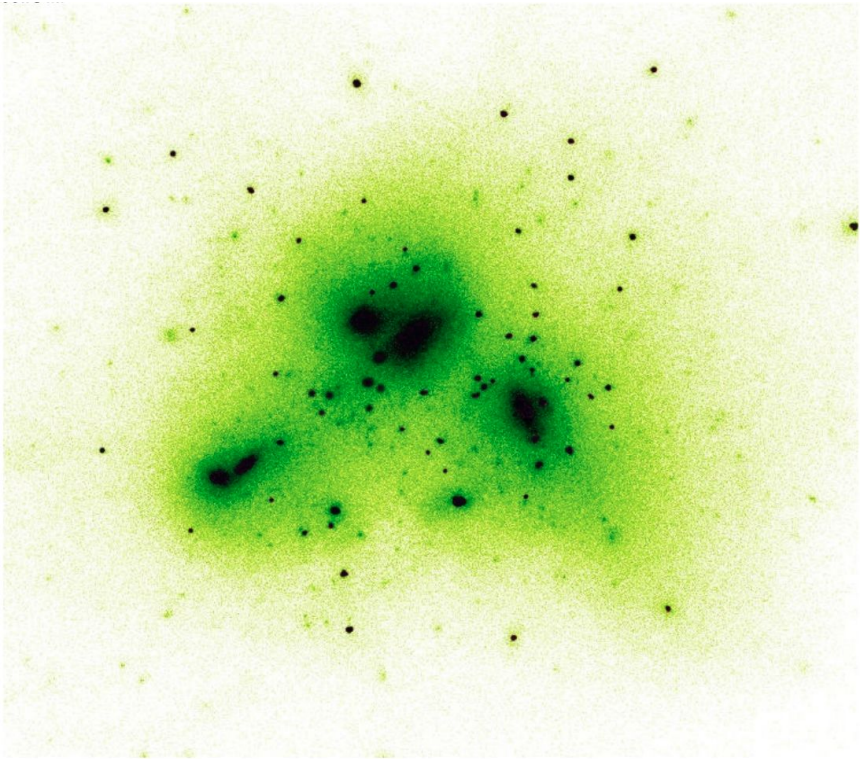
Raw output

- **Full snapshots:** 3D “photographs” of the universe at different times
- **Particle data:** detailed information for dark matter, gas and stars



Post-processing

- **Group/halo catalogs:** identifying and characterising the structure



Credits: AHF user's guide

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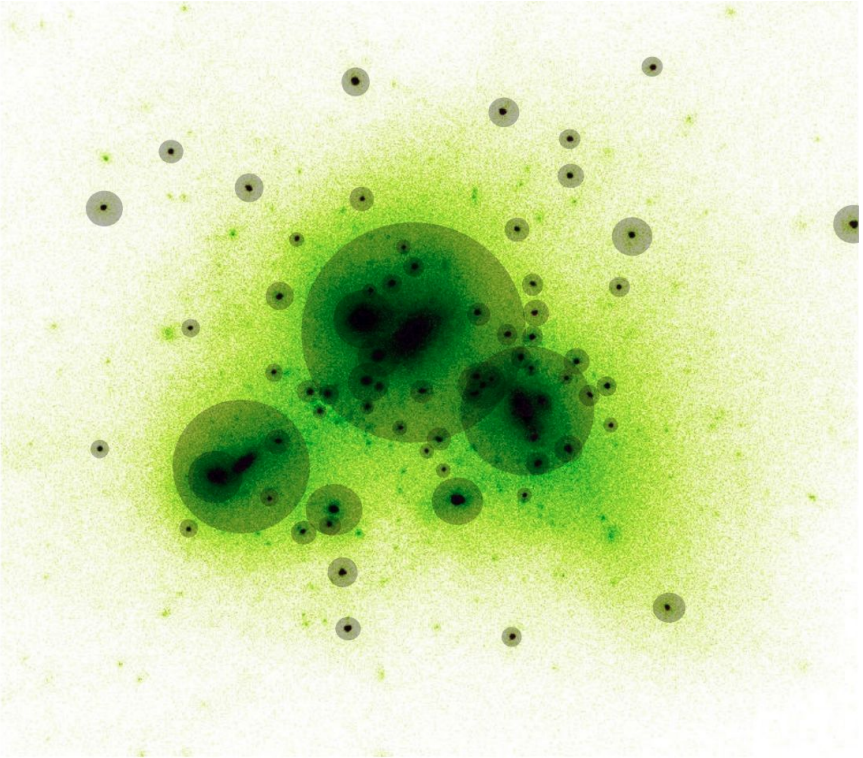

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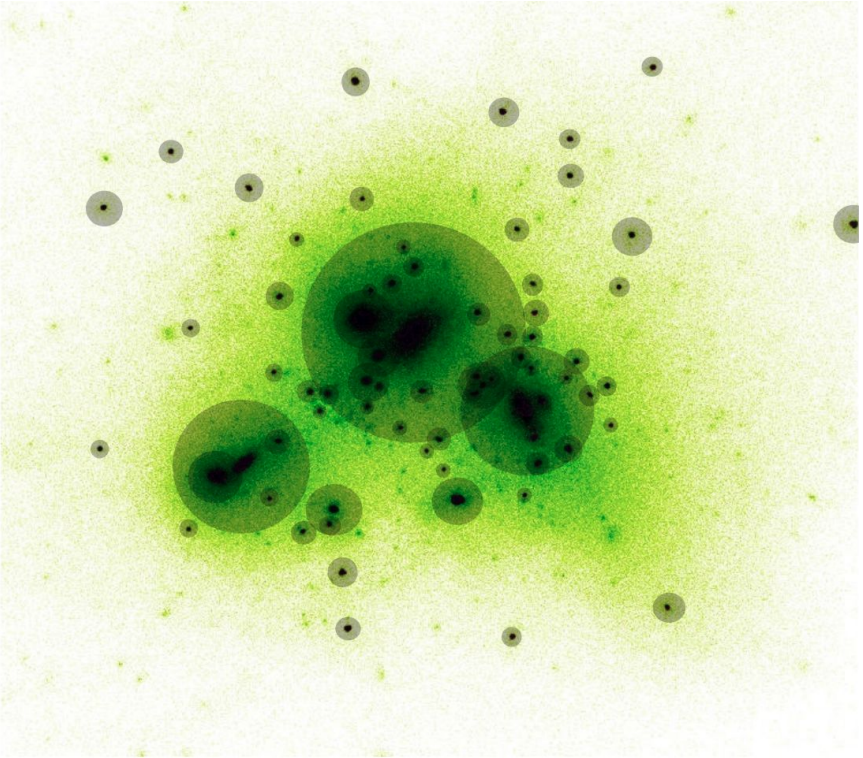
Post-processing

- **Group/halo catalogs:** identifying and characterising the structure

Name	Method	Used by
FoF (Friends -of-Friends)	Links particles within a fixed distance	Base step in most simulations
SUBFIND	Finds gravitationally bound structures within FoF groups	IllustrisTNG, EAGLE
AHF	Adaptive density-peak finder	The Three Hundred
Rockstar	Phase-space based	Bolshoi, MultiDark
VELOCIraptor	Phase-space based	FLAMINGO
HBT+	Time-domain tracking	COLIBRE



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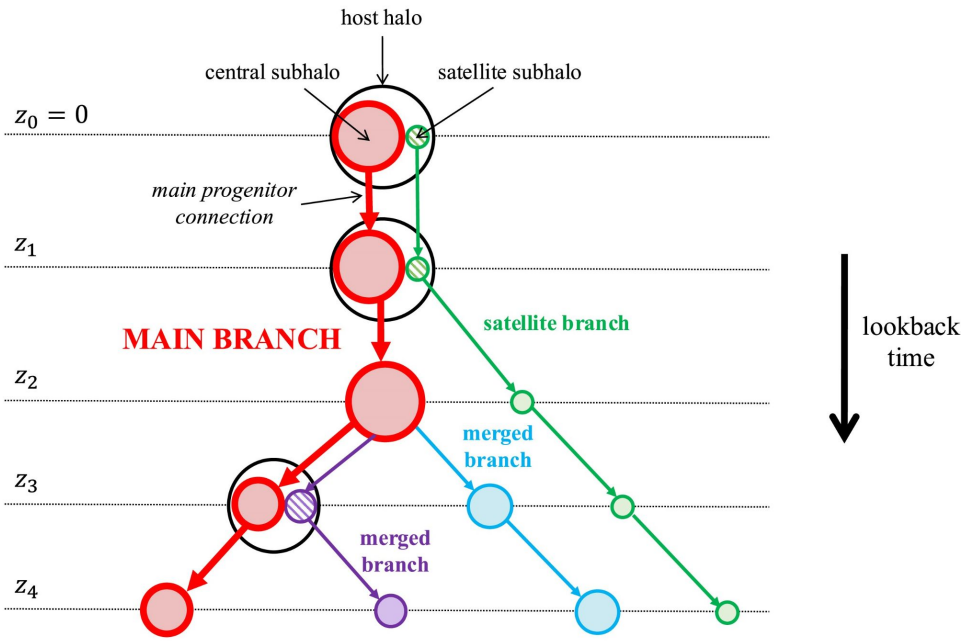
Post-processing

- **Group/halo catalogs:** identifying and characterising the structure
- **Merger trees:** link objects across snapshots



Credits: Chandro-Gómez et al. (2025)

Post-processing

- **Group/halo catalogs:** identifying and characterising the structure
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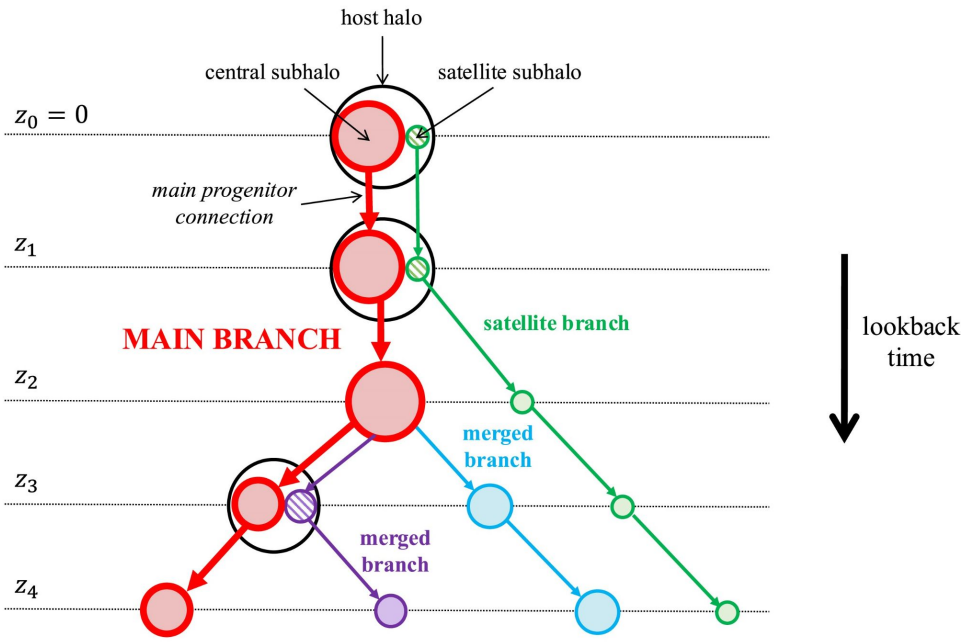
Name	Method	Used by
SUBLINK	Particle overlap + merit function	IllustrisTNG (with SUBFIND)
MergerTree	Particle overlap	With AHF
ConsistentTrees	Particle overlap + temporal consistency	With Rockstar
HBT+	Forward subhalo tracking	COLIBRE



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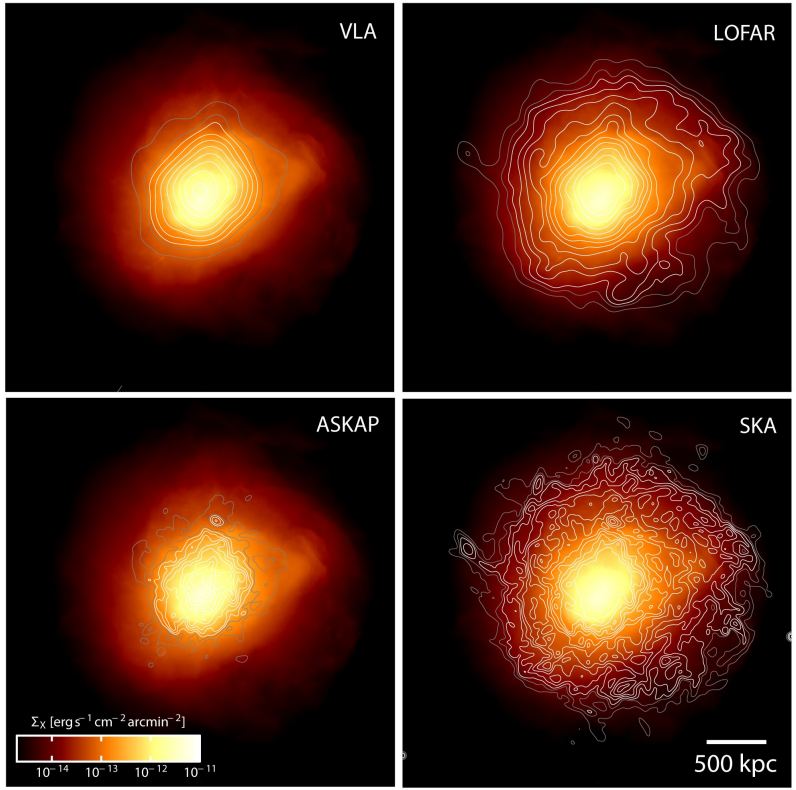
Post-processing

- **Group/halo catalogs:** identifying and characterising the structure
- **Merger trees:** link objects across snapshots
- **Other products:** synthetic images, observables, etc.



X-ray map overlayed with synchrotron emission contours for the most massive halo of the TNG300 simulation. Credits: The TNG Project (Marinacci et al. 2018)

This session

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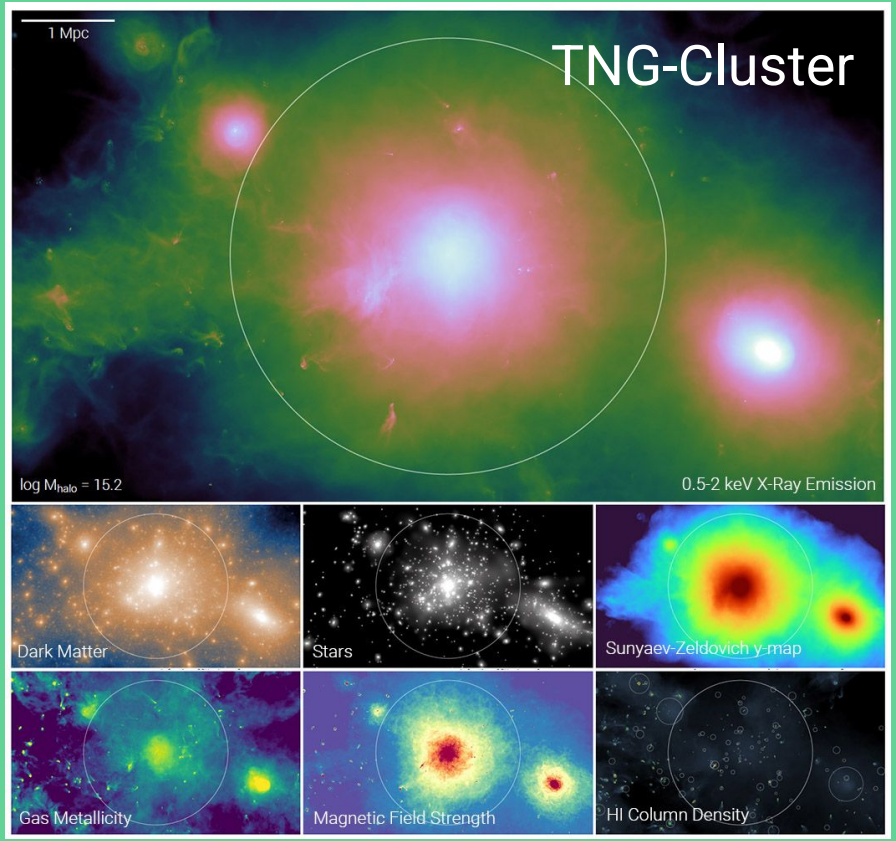
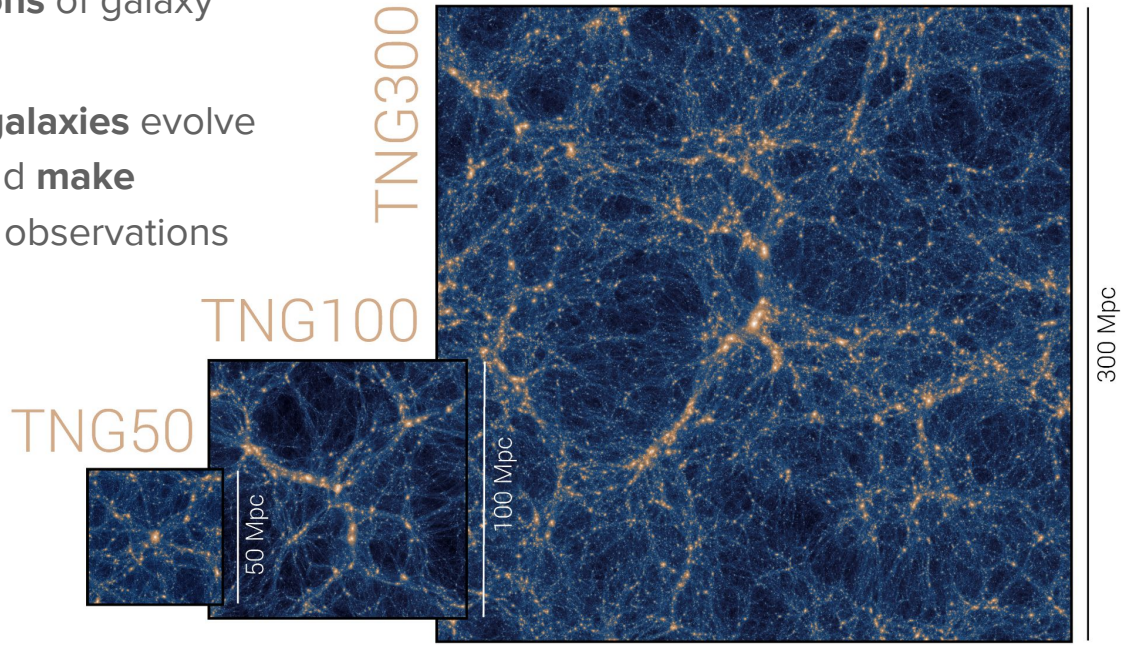


Dylan Nelson (Institute for Theoretical Astrophysics, Heidelberg University)
Annalisa Pillepich (Max Planck Institute for Astronomy, MPIA, Heidelberg)

The TNG-Project

- IllustrisTNG is a series of **large, cosmological magnetohydrodynamic simulations** of galaxy formation
- Aim: **understand** when and how **galaxies** evolve into the structures we observe, and **make predictions** for current and future observations
- Started in 2014, three boxes finished in 2019
- More than 1300 published papers since 2017
- Several spin-off projects, including **TNG-Cluster**



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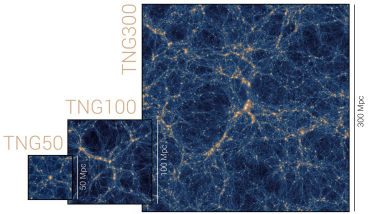
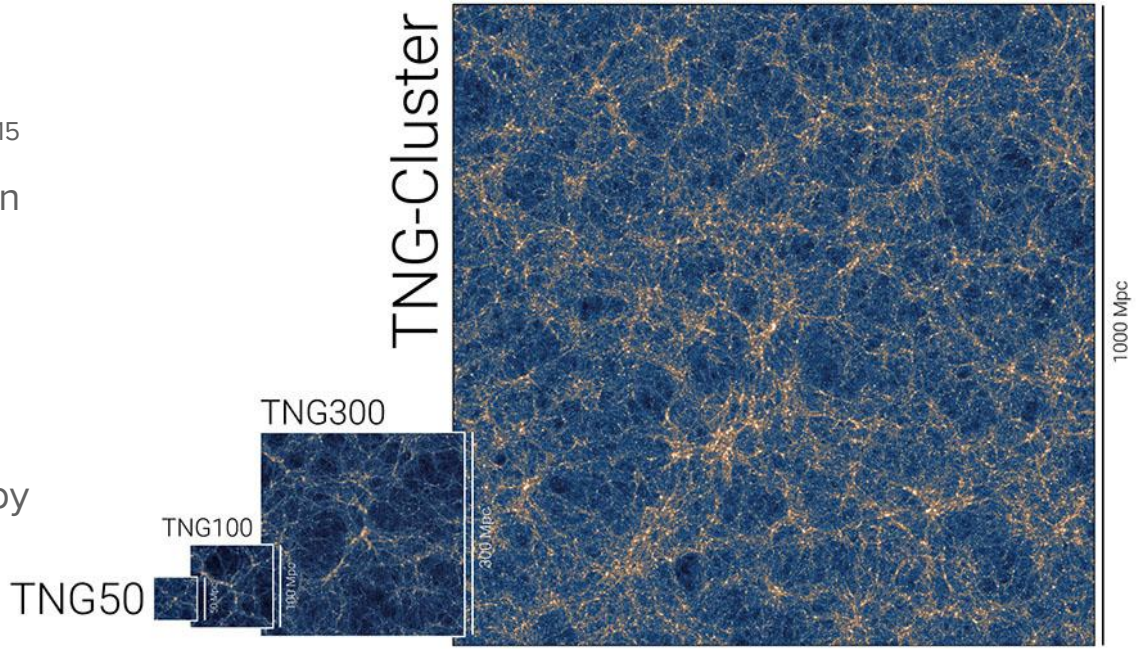
Run		TNG50	TNG100	TNG300
Volume	$[\mathrm{Mpc}^3]$	51.7^3	110.7^3	302.6^3
L_{box}	$[\mathrm{Mpc}/h]$	35	75	205
N_{GAS}	–	2160^3	1820^3	2500^3
N_{DM}	–	2160^3	1820^3	2500^3
N_{TR}	–	2160^3	2×1820^3	2500^3
m_{baryon}	$[M_\odot]$	8.5×10^4	1.4×10^6	1.1×10^7
m_{DM}	$[M_\odot]$	4.5×10^5	7.5×10^6	5.9×10^7
$\epsilon_{\mathrm{gas,min}}$	[pc]	74	185	370
$\epsilon_{\mathrm{DM,stars}}$	[pc]	288	740	1480
N_{groups}	$10^{13.0-14.0}$	23	168	3,545
N_{clusters}	$10^{14.0-14.5}$	1	11	239
N_{clusters}	$10^{14.5-15.0}$	0	3	38
N_{clusters}	$>10^{15.0}$	0	0	3



TNG-Cluster

- Increase the statistical sampling of the most massive and rare objects in the Universe: **massive galaxy clusters**
- **Select** all objects with $M_{\rm halo}/M_{\odot} \gtrsim 10^{15}$ from a parent box of side 1 Gpc run with **dark matter only**
- **Resimulate** them including **hydrodynamics** with the same effective resolution as TNG-300
- **Reconstruct** the full “virtual box” by stitching together all individual simulations



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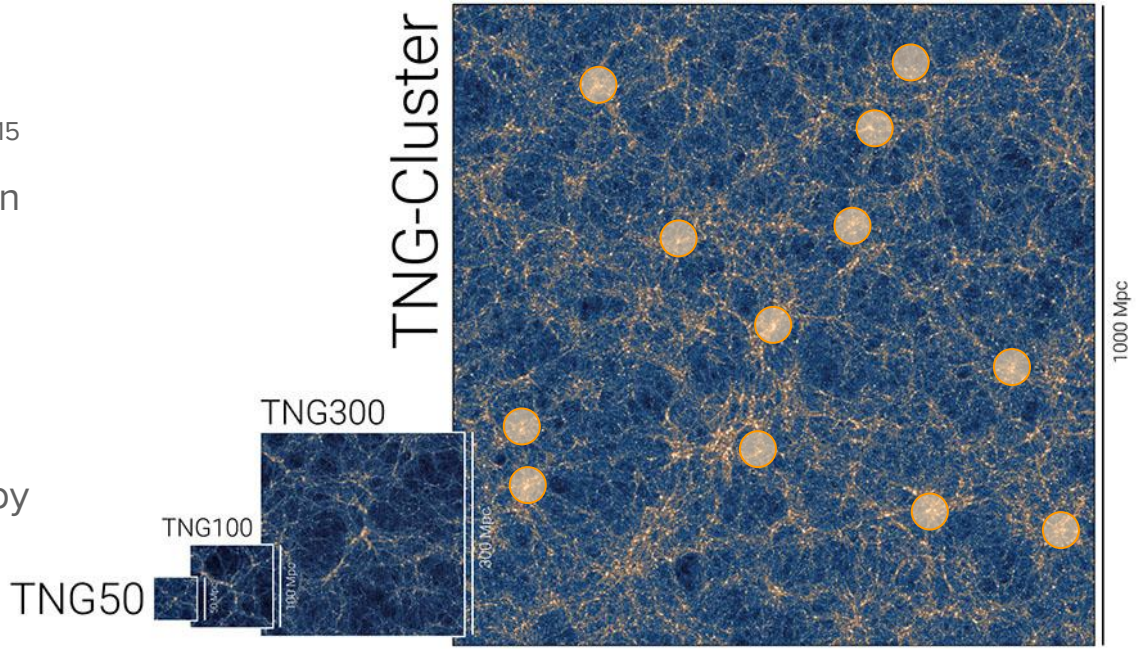

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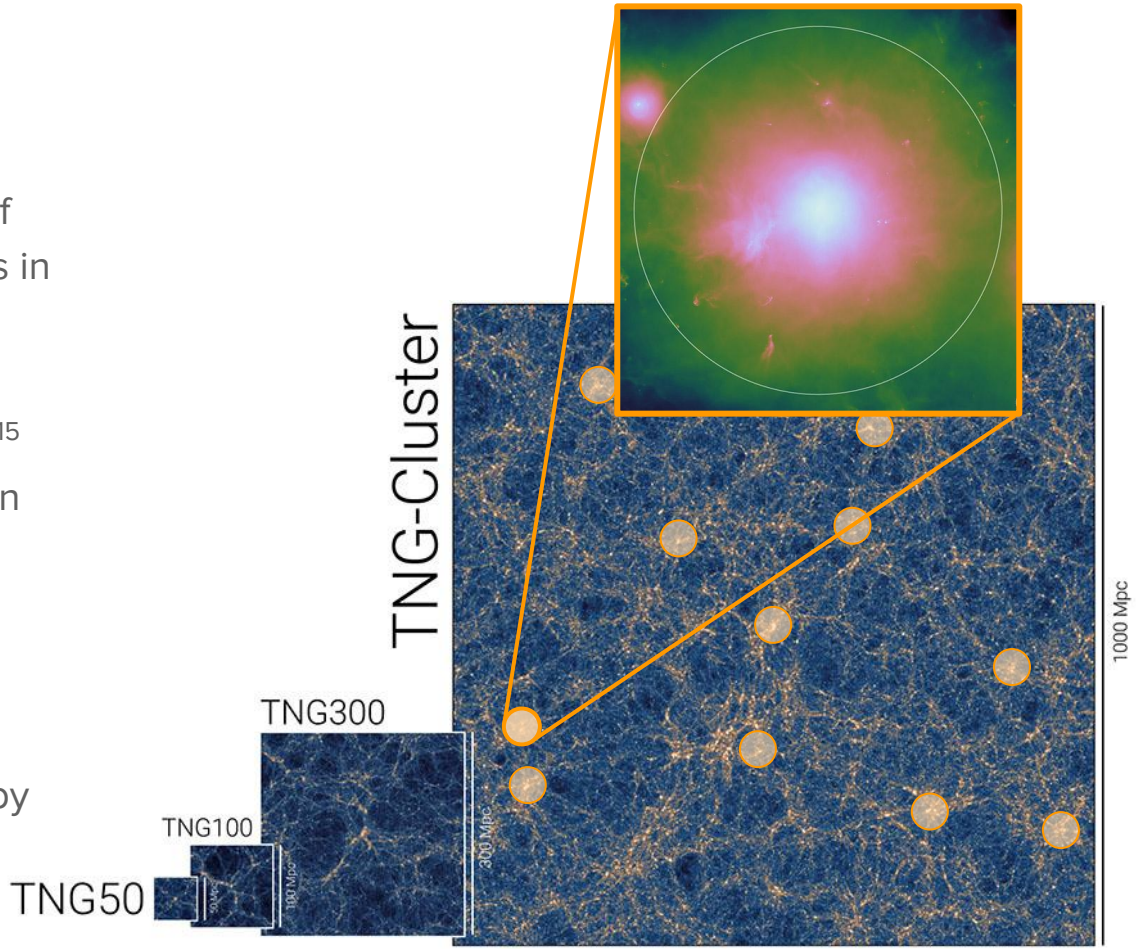

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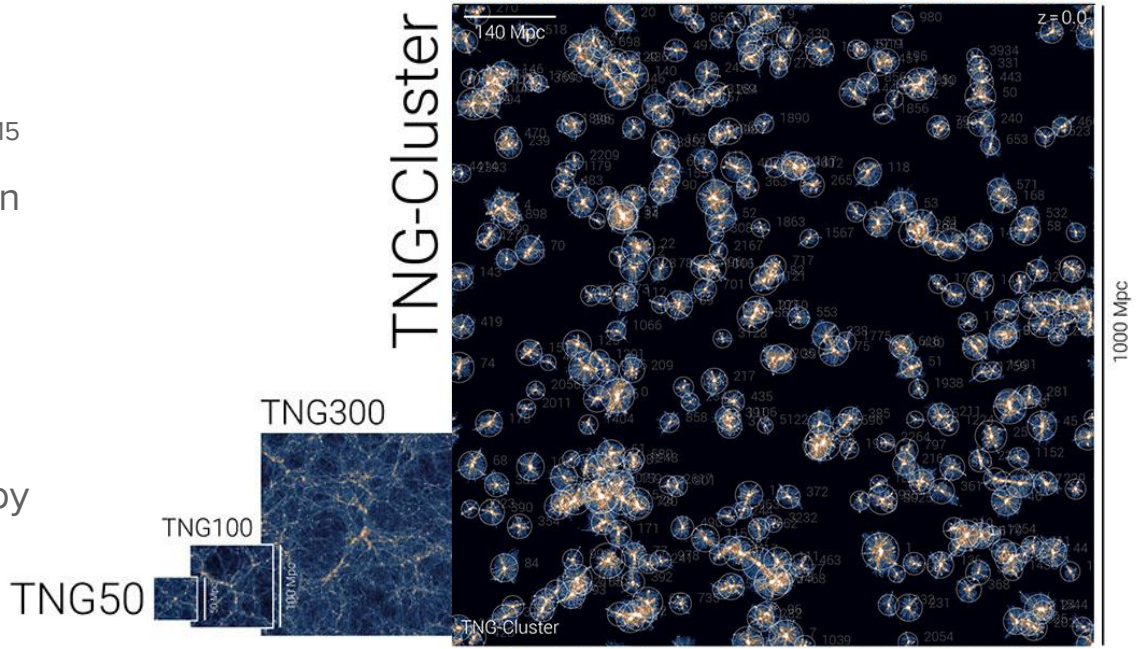
DM: high resolution only (Nelson et al. 2023)



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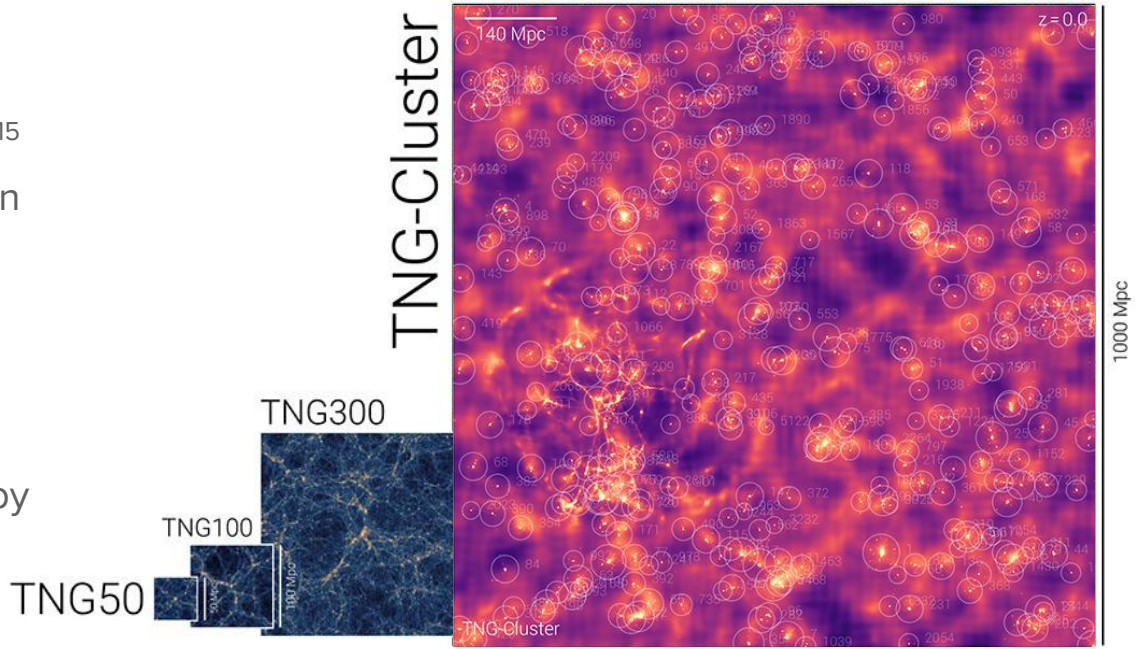
All gas (Nelson et al. 2023)



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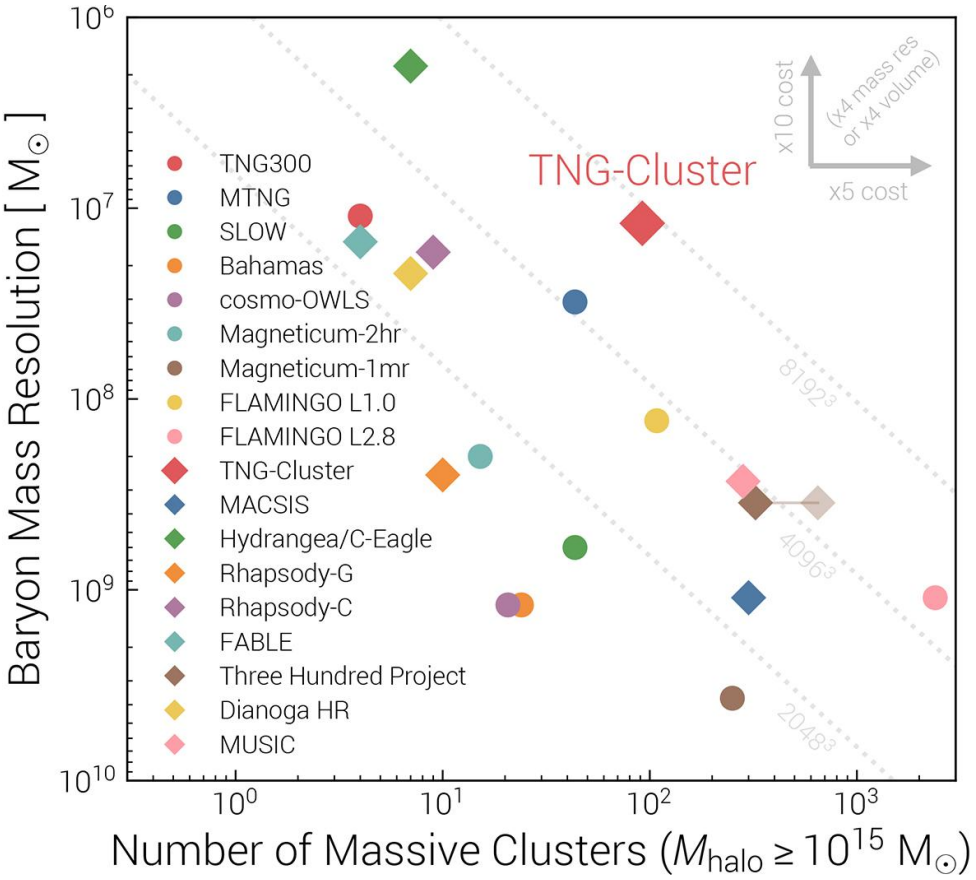
TNG-Cluster
x10 cost
(x4 mass res or x4 volume)
x5 cost
8192³
4096³
2048³
TNG300
MTNG
SLOW
Bahamas
cosmo-OWLS
Magneticum-2hr
Magneticum-1mr
FLAMINGO L1.0
FLAMINGO L2.8
TNG-Cluster
MACSIS
Hydrangea/C-Eagle
Rhapsody-G
Rhapsody-C
FABLE
Three Hundred Project
Dianoga HR
MUSIC
Baryon Mass Resolution [M⊙]
Number of Massive Clusters (Mhalo ≥ 10^15 M⊙)

What we will do in the notebook

TNG-Cluster:

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- Select clusters of interest (massive halos)
- Inspect particle data: selecting the ICL
- Compute observable quantities
- Explore merger trees

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Let’s get to work!